sonic pulses are received by the receiver. To calculate the distance, the Ultrasonic library measures the time required for receiving the pulse and calculates the distance of the object in front of it.

Connect the ultrasonic sensor to the Arduino™ board as shown in the figure.

The ultrasonic sensor uses two measurement units to measure distance, namely 'centimetre' and 'inch'.

An Ultrasonic sensor provides more accurate readings when the distance from object in front is 30 cms to 150 cms.

UNIT FOR MEASUREMENT	SYNTAX
Centimetre	ultrasonic. Ranging (CM)
Inch	ultrasonic. Ranging (ICN)

Using an accelerometer

An accelerometer measures acceleration that any object faces when there's motion or change in motion of the object. It's a very sensitive device and can be used when sensitivity in motion

#define xacc_pin A0;	
#define yacc_pin A1;	
#define zacc_pin A2;	
int zero_point = 290;	Enter zero point for the accelerometer. Check above instruction for setting the zero point.
void setup (){	
Serial.begin (9600); }	Start serial communication at 9.6 kbps
void loop (){	
int xAcc = analogRead (xacc_pin);	Read acceleration along X axis
int yAcc = analogRead (yacc_pin);	Read acceleration along Y axis
int zAcc = analogRead (zacc_pin);	Read acceleration along Z axis
Serial.print ("xAcc ");	
Serial.println (zero_point - xAcc);	Print actual acceleration along X axis
Serial.print ("yAcc ");	
Serial.println (zero_point - yAcc);	Print actual acceleration along Y axis
Serial.print ("zAcc ");	
Serial.println (zero_point - zAcc);	Print actual acceleration along Z axis
Serial.println (" ");	
delay (250); }	Wait for 250 milliseconds or th of a second